

Elite Full-Stack Developer Roadmap (2026+)

Bootcamp-grade curriculum designed to take a learner from foundations to production-ready full-stack engineering.

Module 0: Computer Science & Engineering Foundations

Estimated Time: 4-6 weeks (80-120 hours)

Granular Checklist

- Computer architecture, operating systems, networking fundamentals
- HTTP, HTTPS, DNS, TCP/IP, browser architecture
- Linux terminal, shell scripting, package managers
- Git internals, branching strategies, rebasing, code reviews
- Big-O analysis, arrays, linked lists, stacks, queues, trees, graphs, hash maps
- Debugging workflows, profiling, logging, AI-assisted development with Copilot/Cursor

Production Projects

- Developer Productivity Platform
- Git Repository Analytics System

Module 1: HTML, CSS & UI Engineering

Estimated Time: 4-6 weeks (100-140 hours)

Granular Checklist

- Semantic HTML5
- Accessibility (WCAG, ARIA)
- Forms and validation
- Flexbox, Grid, responsive design
- CSS architecture (BEM, utility-first)
- Tailwind CSS
- Animations, rendering pipeline, performance optimization
- Design systems and component libraries

Production Projects

- Enterprise Design System
- Multi-brand Marketing Platform

Module 2: JavaScript Deep Dive

Estimated Time: 8-10 weeks (180-250 hours)

Granular Checklist

- Execution context
- Scope chain and closures

- Prototypes and inheritance
- this binding
- Event loop, microtasks, macrotasks
- Promises, async/await
- Fetch API
- Modules
- Memory management and garbage collection
- Functional programming concepts
- Advanced debugging

Production Projects

- Real-time Collaboration Engine
- Workflow Automation Tool

Module 3: TypeScript

Estimated Time: 3-5 weeks (80-120 hours)

Granular Checklist

- Static typing
- Interfaces
- Type aliases
- Generics
- Utility types
- Mapped types
- Conditional types
- Type narrowing
- Large-scale architecture patterns

Production Projects

- Typed SDK Generator
- Validation Framework

Module 4: React Engineering

Estimated Time: 8-10 weeks (180-250 hours)

Granular Checklist

- JSX
- Components
- Props and state
- Hooks

- Context API
- Rendering lifecycle
- Reconciliation
- Performance optimization
- Code splitting
- Testing
- State management with Zustand/TanStack Query

Production Projects

- SaaS Analytics Dashboard
- Real-time Team Workspace

Module 5: Next.js

Estimated Time: 6-8 weeks (150-220 hours)

Granular Checklist

- App Router
- Server Components
- Server Actions
- SSR
- SSG
- ISR
- Caching
- SEO
- Authentication
- Observability
- Deployment patterns

Production Projects

- AI Knowledge Platform
- Enterprise E-commerce System

Module 6: Backend Engineering

Estimated Time: 8-10 weeks (180-250 hours)

Granular Checklist

- Node.js runtime internals
- Express or NestJS
- REST APIs
- Validation

- Authentication
- Authorization
- JWT
- OAuth
- Logging
- Error handling
- Queues
- Background jobs
- API versioning
- Rate limiting

Production Projects

- Booking Platform Backend
- Notification Service

Module 7: Database Engineering

Estimated Time: 6-8 weeks (140-200 hours)

Granular Checklist

- PostgreSQL
- Schema design
- Normalization
- Transactions
- Indexes
- Query planning
- Prisma ORM
- Redis caching
- Replication
- Backups
- Migrations

Production Projects

- Financial Ledger System
- Analytics Warehouse

Module 8: Testing & Quality

Estimated Time: 3-5 weeks (80-120 hours)

Granular Checklist

- Unit testing

- Integration testing
- E2E testing
- Jest
- Vitest
- Playwright
- Contract testing
- Coverage analysis
- CI quality gates

Production Projects

- Automated QA Platform
- Regression Framework

Module 9: DevOps & Cloud

Estimated Time: 6-8 weeks (140-220 hours)

Granular Checklist

- Docker
- Containerization
- GitHub Actions
- AWS fundamentals
- Infrastructure as Code
- Terraform
- Monitoring
- Logging
- Secrets management
- Deployment strategies

Production Projects

- Production CI/CD Pipeline
- Cloud SaaS Infrastructure

Module 10: System Design

Estimated Time: 6-8 weeks (120-180 hours)

Granular Checklist

- CAP theorem
- Caching
- Load balancing
- Message queues

- Consistency models
- Sharding
- Distributed systems
- Scalability reviews
- Observability

Production Projects

- Messaging Platform Architecture
- Video Platform Architecture

Module 11: AI-Augmented Engineering

Estimated Time: 2-4 weeks (40-80 hours)

Granular Checklist

- Prompt engineering
- LLM APIs
- Embeddings
- RAG architecture
- Vector databases
- Agents
- Evaluation pipelines
- Secure AI integration

Production Projects

- Enterprise AI Copilot
- Knowledge Assistant

Capstone Portfolio Requirements

- Enterprise SaaS Platform with authentication, billing, analytics, CI/CD and observability.
- AI-enabled product using Next.js, TypeScript, PostgreSQL, Redis and cloud deployment.
- Document architecture decisions, testing strategy, scalability plan and monitoring approach.
- Deploy production environments and demonstrate rollback procedures.